Introduction to Complex Ions

- A **complex** is formed when a central metal atom or ion is surrounded by other molecules or ions which form **dative covalent bonds** with the central metal atom or ion.
- The molecules or ions that form the dative covalent bonds are called **ligands**
- In a ligand, there is at least one atom having **a lone pair of electrons** which can be donated to the central metal atom or ion to form a dative covalent bond
- Depending on the overall charge of the complex formed, complexes are classified into 3 main types:

i. **cationic complex** (e.g. $[Cr(NH_3)_6]^{3+}$, $[Zn(NH_3)_4]^{2+}$)

ii. **neutral complex** (e.g. $Co(NH_3)_3Cl_3$)

iii. **anionic complex** (e.g. $[Fe(CN)_6]^{3-}$, $[CoCl_4]^{2-}$)

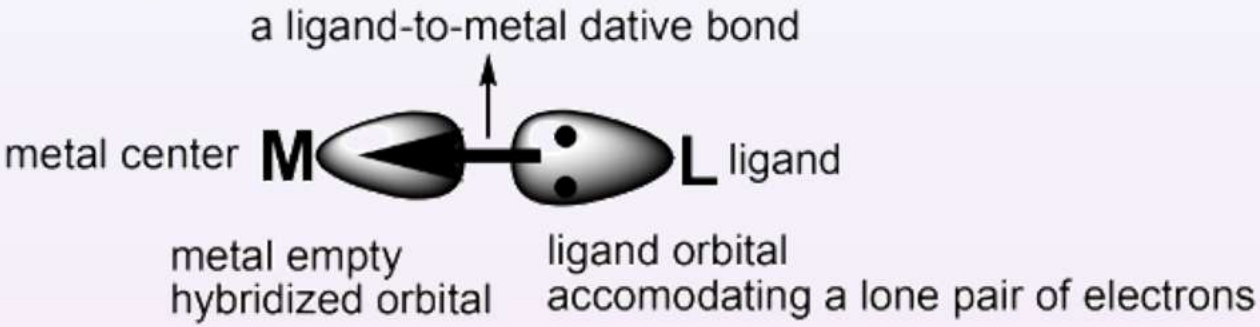
Introduction to Complex Ions

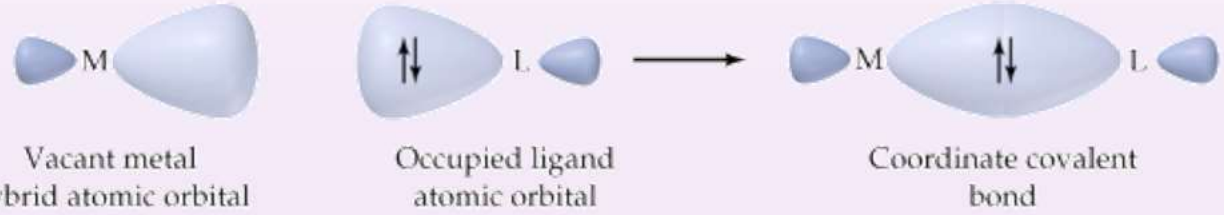
Formation of Complexes

- ***Coordination compounds*** are substances containing complex ions.
- The **coordination number** of the central metal atom or ion in a complex is the *number of ligands bonded to this metal atom or ion.*
- The most common coordination numbers are **4** and **6.**
- Other coordination numbers are still possible, for example, Ag can form a complex with coordination number of 2. (e.g. $[Ag(NH_3)_2]^+$)
- For the first series of *d*-block metals, complexes are formed using the **3*d*, 4*s*, 4*p* and 4*d* orbitals** present in the metal atoms or ions. Due to the presence of *vacant, low energy orbitals*, *d*-block metals can interact with the orbitals of the surrounding ligands
- Due to the relatively *small sizes and high charge* of *d*-block metal ions, they introduce *strong polarization* on the ligands. This *favours the formation of bonds of high covalent character*.

Introduction to Complex Ions

Formation of Complexes





$[CoF_6]^{3-}$: [Ar] ↑↓ ↑ ↑ ↑ ↑ | ↑↓ ↑↓ ↑↓ ↑↓ ↑↓ ↑↓ | — — —

3*d* 4*s* 4*p* 4*d*

(paramagnetic)

Six sp^3d^2 bonds to the ligands

$[Co(CN)_6]^{3-}$: [Ar] ↑↓ ↑↓ ↑↓ | ↑↓ ↑↓ ↑↓ ↑↓ ↑↓ ↑↓ | — — — — —

3*d* 4*s* 4*p* 4*d*

(diamagnetic)

Six d^2sp^3 bonds to the ligands

Introduction to Complex Ions

Nature of Ligands

- Ligands that produce a *larger splitting* are called **strong field ligands**, often results in *low-spin complex*, and those that produce a *smaller splitting* are called **weak field ligands**, often results in *high-spin complex*.

Some ligands in spectrochemical series

Weak field ligand: $I^- < Br^- < Cl^- < \underline{S}CN^- < NO_3^- < F^- < OH^- < C_2O_4^{2-} < \underline{O}NO^- < H_2O < \underline{S}CN^- < EDTA^{4-} < py < NH_3 < en < \underline{N}O_2^- < PPh_3 < CN^- < CO$: *Strong field ligand*

Nomenclature of Complexes (No. of ligand + ligand prefix + metal suffix + O.N.)

Metal	*Name in anionic complex
Titanium	Titanate
Vanadium	Vanadate
Chromium	Chromate
Manganese	Manganate
Iron	Ferrate
Cobalt	Cobaltate
Nickel	Nickelate
Copper	Cuprate
Zinc	Zincate
Platinum	Platinate

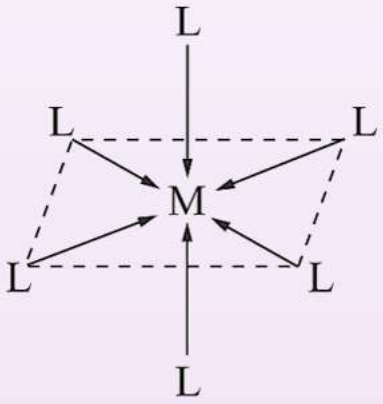
Ligand	Name of ligand
Bromide (Br^-)	Bromo
Chloride (Cl^-)	Chloro
Fluoride (F^-)	Fluoro
Hydroxide (OH^-)	Hydroxo
Water (H_2O)	Aqua
Ammonia (NH_3)	Ammine
Nitrite (NO_2^-)	Nitrito-*N*
Cyanide (CN^-)	Cyano
Carbon monoxide (CO)	Cabonyl

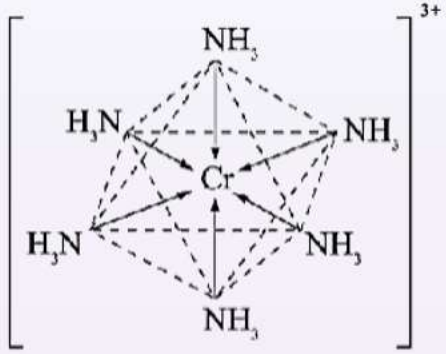
*Metal in **cationic complex** remains unchanged

Introduction to Complex Ions

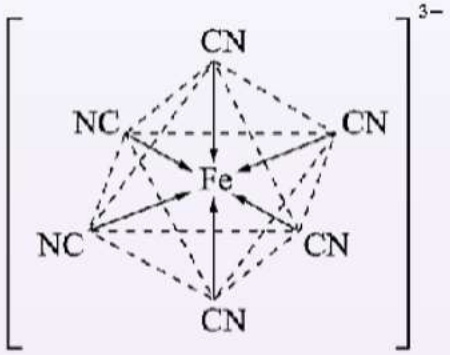
Shape of Complexes

1. Octahedral

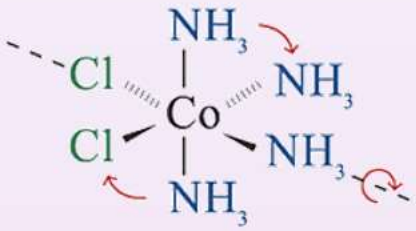




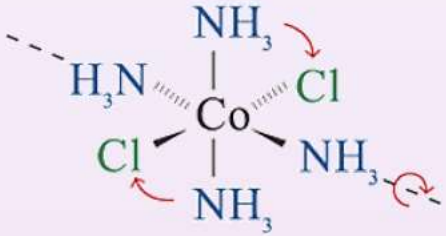
Hexaamminechromium(III) ion, $[Cr(NH_3)_6]^{3+}$



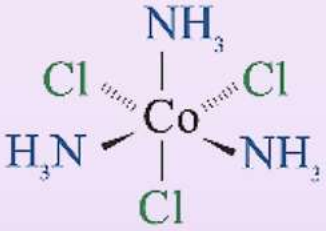
Hexacyanoferrate(III) ion, $[Fe(CN)_6]^{3-}$



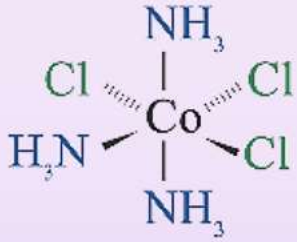
cis-dichlorotetraamminecobalt(III) ion



trans-dichlorotetraamminecobalt(III) ion



fac-trichlorotriamminecobalt(III)

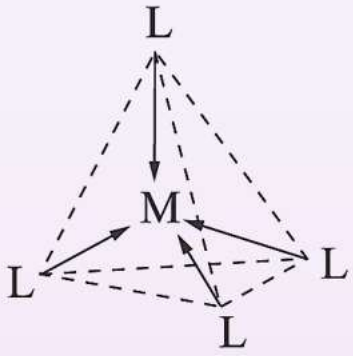


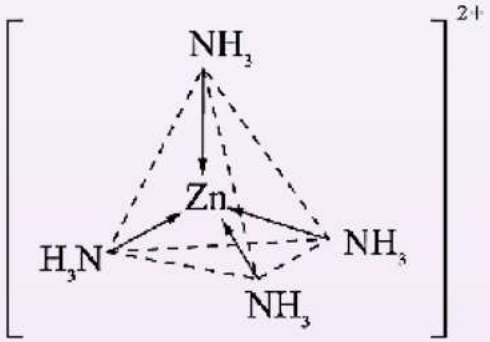
mer-trichlorotriamminecobalt(III)

Introduction to Complex Ions

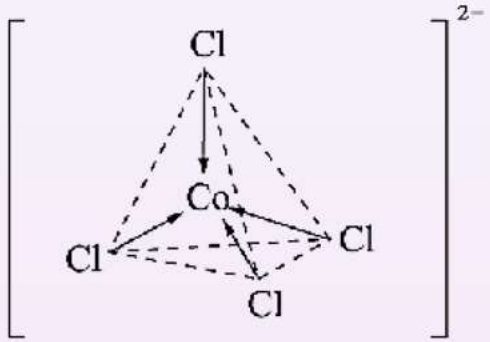
Shape of Complexes

2. Tetrahedral



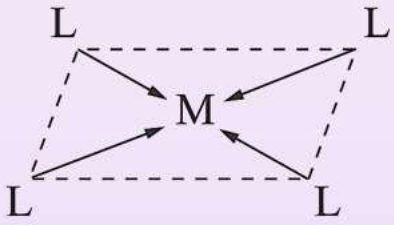


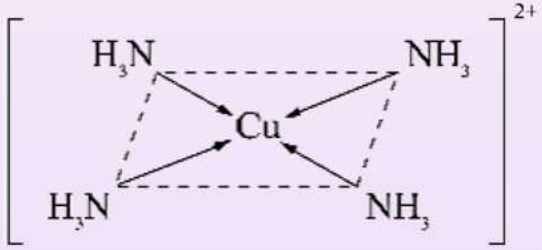
Tetraamminezinc(II) ion, $[Zn(NH_3)_4]^{2+}$



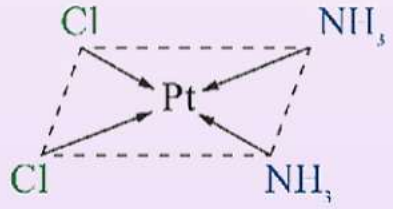
Tetrachlorocobaltate(II) ion, $[CoCl_4]^{2-}$

3. Square planar





Tetraamminecopper(II) ion, $[Cu(NH_3)_4]^{2+}$

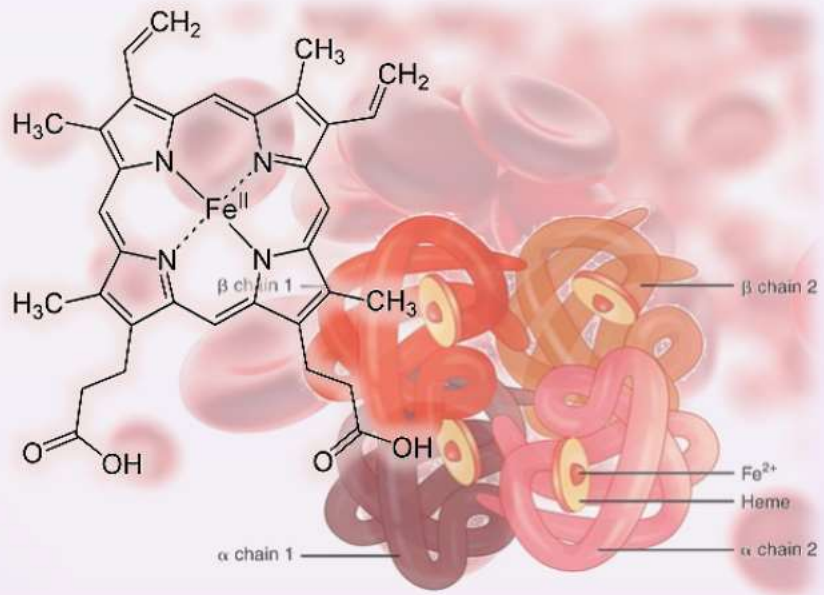


cis-Diamminedichloroplatinum(II), [*cis*-$[Pt(NH_3)_2Cl_2]$ (cisplatin)

Importance of Complex Ions

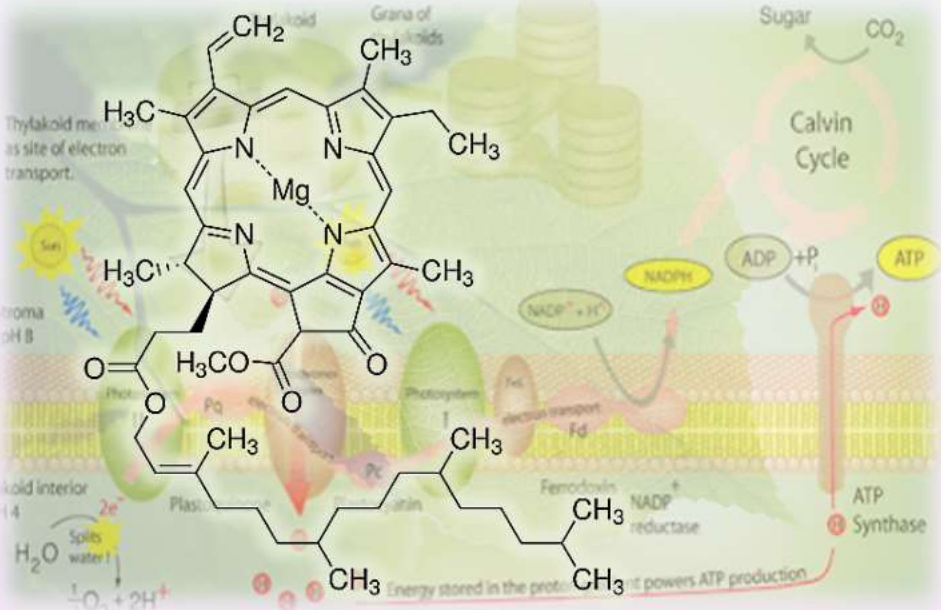
Metal ions as a centre coordinated with heterocyclic ligands

Porphyrin ligand



Iron (II) ion coordinated with porphyrin ligand to form Heme B – which is coordinated with globular subunit of haemoglobin, the iron (II) complex could further coordinate with (carry) an oxygen molecule and transport it in blood.

Chlorin ligand



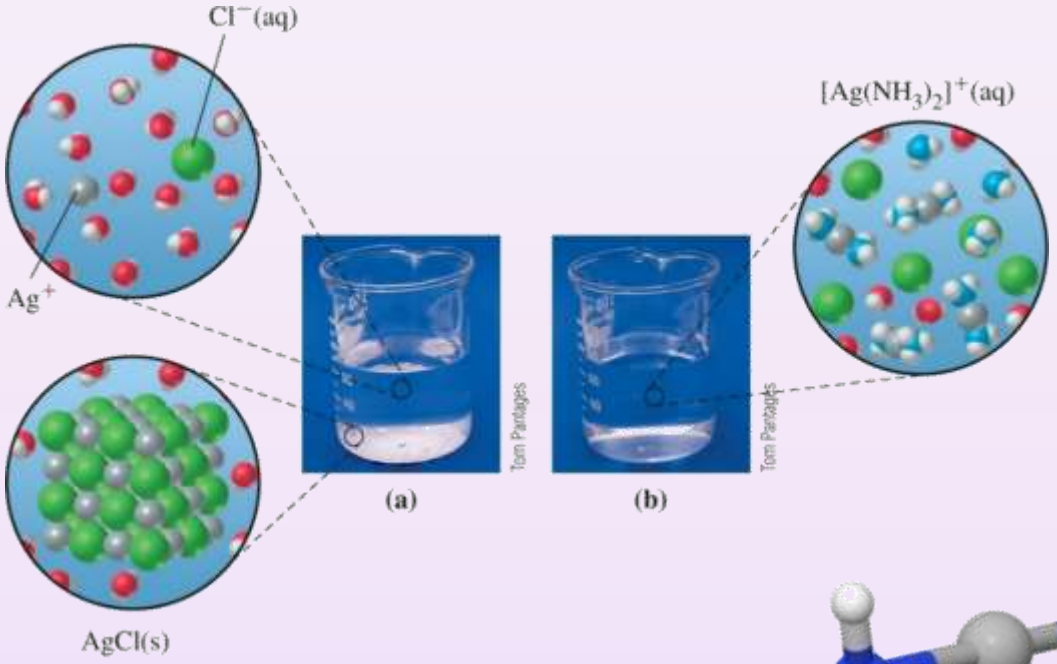
Magnesium ion is coordinated with chlorin ligand to form chlorophyll *a* – which mediate the transfer of electrons in photosystem of electron transport chain in the light dependent stage

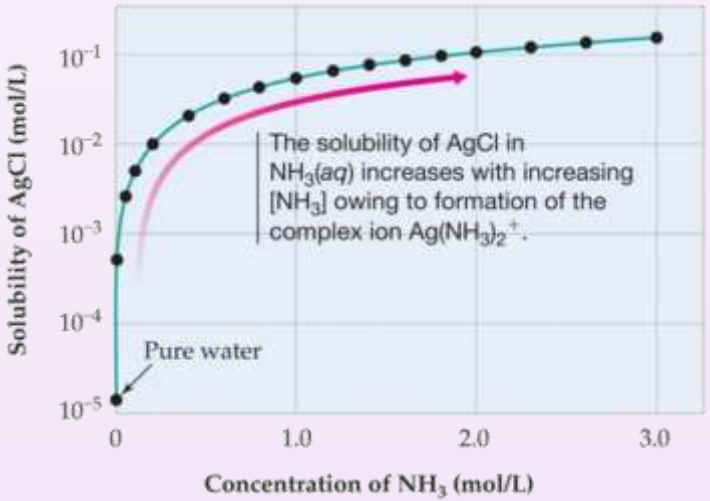
Equilibria Involving Complex Ions

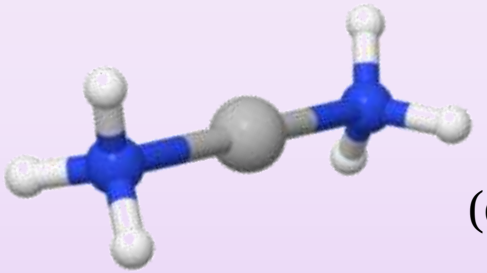
Example: Formation of $[Ag(NH_3)_2]^+$

Consider excess AgCl precipitates in pure water, when $NH_3(aq)$ is added, Ag^+ ions from AgCl combine with NH_3 molecules to form the ions **$[Ag(NH_3)_2]^+$** which, together with Cl^- ions, remain in solution as the soluble compound $Ag(NH_3)_2Cl$.

$$AgCl(s) + 2\,NH_3(aq) \longrightarrow [Ag(NH_3)_2]^+(aq) + Cl^-(aq) \qquad (4)$$







$[Ag(NH_3)_2]^+$
(diamminesilver(I) cation)

Equilibria Involving Complex Ions

Example: Formation of $[Ag(NH_3)_2]^+$

- Reaction (4) may be thought as involving **two** simultaneous equilibria.

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq) \quad (5)$$

$$Ag^+(aq) + 2\,NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+(aq) \quad (6)$$

$$AgCl(s) + 2\,NH_3(aq) \longrightarrow [Ag(NH_3)_2]^+(aq) + Cl^-(aq) \quad (4)$$

- The equilibrium in reaction (6) is shifted far to the right because $[Ag(NH_3)_2]^+$ is a stable complex ion, so $[Ag^+]_{eqm}$ is kept so low that the ion product $[Ag^+][Cl^-] < K_{sp}$. $\therefore$ *AgCl remains in solution*.

- The equilibrium constant that is used to deal with a complex-ion equilibrium is called the ***formation (stability) constant***, K_f. This describes the formation of a complex ion from a central ion and its attached groups.

Equilibria Involving Complex Ions

- For reaction (6) this equilibrium constant expression is

$$K_f = \frac{[[Ag(NH_3)_2]^+]}{[Ag^+][NH_3]^2} = 1.6 \times 10^7$$

$$Ag^+(aq) + 2\,NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+(aq) \qquad (6)$$

- Reaction (6) takes place in ***two*** steps:

(step 1) $Ag^+(aq) + NH_3(aq) \rightleftharpoons Ag(NH_3)^+(aq)$ $\quad \frac{[Ag(NH_3)^+]}{[Ag^+][NH_3]} = K_1 = 2.1 \times 10^3$

(step 2) $Ag(NH_3)^+(aq) + NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+(aq)$ $\quad \frac{[Ag(NH_3)_2^+]}{[Ag(NH_3)^+][NH_3]} = K_2 = 8.2 \times 10^3$

Adding the above two equations, reaction (6) is obtained.

$$K_f = K_1K_2 = \frac{[Ag(NH_3)_2^+]}{[Ag^+][NH_3]^2}$$

Example 9
Predicting Reactions Involving Complex Ions

Predict what will happen if nitric acid is added to a solution of $[Ag(NH_3)_2]Cl$ in NH_3 (aq).



Reprecipitating AgCl(s) The reagent being added to the solution containing $[Ag(NH_3)_2]^+$ and Cl^- is $HNO_3(aq)$. H_3O^+ from the acid reacts with $NH_3(aq)$ to form $NH_4^+(aq)$. As a result, equilibrium between $[Ag(NH_3)_2]^+$, Ag^+, and NH_3 is upset. The complex ion is destroyed, $[Ag^+]$ quickly rises to the point at which K_{sp} for AgCl is exceeded, and a precipitate forms.

Suggested Solution:

Since $HNO_3(aq)$ is a strong acid, we will represent the acid as $H_3O^+(aq)$ and write the protonation reaction as

$$H_3O^+(aq) + NH_3(aq) \longrightarrow NH_4^+(aq) + H_2O(l)$$

The formation reaction is

$$Ag^+(aq) + 2\,NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+(aq)$$

To replace free NH_3 lost in this neutralization, equilibrium in the formation reaction shifts to the *left*. As a result, $[Ag^+]$ increases. When $[Ag^+]$ increases to the point at which the ion product $[Ag^+][Cl^-]$ exceeds K_{sp}, AgCl(s) precipitates

Substitution of Ligands in Complex Ions

- Ligand substitution involves the exchange of one ligand for another, with no change in oxidation state at the metal centre.
- Example: The substitutions of the H_2O ligands in $[M(H_2O)_m]$ by another ligand L can be generally represented as:

$$\mathbf{[M(H_2O)_{\mathit{m}}] + \mathit{m}L \rightleftharpoons [ML_{\mathit{m}}] + \mathit{m}H_2O} \quad \textit{(charge ignored)}$$

- The formation constant (K_f) for the complex $[ML_m]$ is:

$$\boldsymbol{K_f} = \frac{[ML_m]}{[M(H_2O)_m][L]^m} \quad \textit{(charge ignored)}$$

- Ligand substitution reactions depend on the *strength of ligands* and the *stability of complex ions*.

- Complex ions are usually **coloured** and the ***colours are often related to the types of ligands present***, which can be followed during experiments easily

Example: $[Fe(H_2O)_6]^{2+}(aq) + 6CN^-(aq) \rightleftharpoons [Fe(CN)_6]^{4-}(aq) + 6H_2O(l)$

hexaaquairon(II) ion — hexacyanoferrate(II) ion

$[Ni(H_2O)_6]^{2+}(aq) + 6NH_3(aq) \rightleftharpoons [Ni(NH_3)_6]^{2+}(aq) + 6H_2O(l)$

hexaaquanickel(II) ion — hexaamminenickel(II) ion

Substitution of Ligands in Complex Ions

Application: $[Fe(CN)_6]^{3-}(aq)$ in industry

$$[Fe(H_2O)_6]^{3+}(aq) + 6CN^-(aq) \rightleftharpoons [Fe(CN)_6]^{3-}(aq) + 6H_2O(l)$$

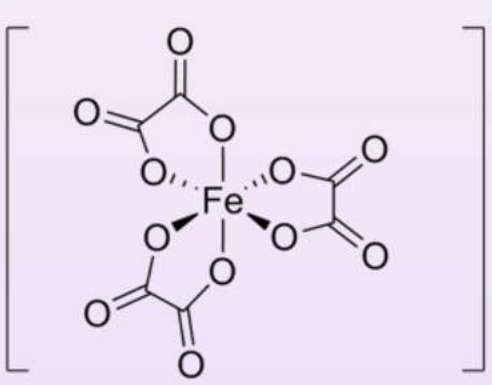
hexaaquairon(III) ion — hexacyanoferrate(II) ion

$$[Fe(CN)_6]^{3-}(aq) + [Fe(C_2O_4)_3]^{3-}(aq) \rightarrow Fe^{3+}{}_4[Fe(CN)_6]^{4-}{}_3(aq) + 6CO_2(g)$$

hexacyanoferrate(II) ion — tris(oxalate)ferrate ion — iron(II,III) hexacyanidoferrate(II,III) / Prussian Blue

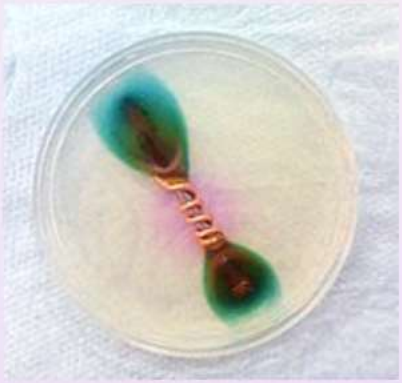
$$[Fe(CN)_6]^{3-}(aq) + [Fe(H_2O)_6]^{2+}(aq) \rightarrow Fe^{3+}{}_4[Fe(CN)_6]^{4-}{}_3(aq) + 6H_2O(l)$$

- The colour change of $[Fe(CN)_6]^{3-}(aq)$ are widely used in *blueprint* in textile industry and *rust indicator* to detect the presence of iron (II) ions.



tris(oxalate)ferrate ion





Substitution of Ligands in Complex Ions

Example 1: Copper (II) ion in excess ammonia solution

- Cations (e.g. Cu^{2+}) in aqueous solution exist mostly in *hydrated* form (e.g. $[Cu(H_2O)_4]^{2+}$). When NH_3 molecules bond to Cu^{2+} and form an ammine complex ion, they do not enter an empty coordination sphere. They must displace H_2O molecules. This displacement occurs in a stepwise fashion.

$$[Cu(H_2O)_4]^{2+}(aq) + NH_3(aq) \rightleftharpoons [Cu(NH_3)(H_2O)_3]^{2+}(aq) + H_2O(l)$$

$$K_1 = 1.9 \times 10^4 \text{ dm}^3\text{mol}^{-1}$$

$$[Cu(NH_3)(H_2O)_3]^{2+}(aq) + NH_3(aq) \rightleftharpoons [Cu(NH_3)_2(H_2O)_2]^{2+}(aq) + H_2O(l)$$

$$K_2 = 3.9 \times 10^3 \text{ dm}^3\text{mol}^{-1}$$

$$[Cu(NH_3)_2(H_2O)_2]^{2+}(aq) + NH_3(aq) \rightleftharpoons [Cu(NH_3)_3(H_2O)]^{2+}(aq) + H_2O(l)$$

$$K_3 = 1.0 \times 10^3 \text{ dm}^3\text{mol}^{-1}$$

$$[Cu(NH_3)_3(H_2O)]^{2+}(aq) + NH_3(aq) \rightleftharpoons [Cu(NH_3)_4]^{2+}(aq) + H_2O(l)$$

$$K_4 = 1.5 \times 10^2 \text{ dm}^3\text{mol}^{-1}$$

$$[Cu(H_2O)_4]^{2+}(aq) + 4NH_3(aq) \rightleftharpoons [Cu(NH_3)_4]^{2+}(aq) + 4H_2O(l)$$

tetraaquacopper(II) ion **tetraamminecopper(II) ion**

$$K_f = K_1 \times K_2 \times K_3 \times K_4 = 1.1 \times 10^{13} \text{ dm}^{12}\text{mol}^{-4}$$

Substitution of Ligands in Complex Ions

Example 2: Copper (II) ion in concentrated hydrochloric acid

- When concentrated HCl is added dropwise, the Cl^- bond to Cu^{2+} and form a metal chloride complex ion in a stepwise fashion.

$$[Cu(H_2O)_4]^{2+}(aq) + Cl^-(aq) \rightleftharpoons [Cu(H_2O)_3Cl]^+(aq) + H_2O(l)$$

$$K_1 = 6.3 \times 10^2 \ dm^3mol^{-1}$$

$$[Cu(H_2O)_3Cl]^+(aq) + Cl^-(aq) \ [Cu(H_2O)_2Cl_2](aq) + H_2O(l)$$

$$K_2 = 4.0 \times 10^1 \ dm^3mol^{-1}$$

$$[Cu(H_2O)_2Cl_2](aq) + Cl^-(aq) \rightleftharpoons [Cu(H_2O)Cl_3]^-(aq) + H_2O(l)$$

$$K_3 = 5.4 \ dm^3mol^{-1}$$

$$[Cu(H_2O)Cl_3]^-(aq) + Cl^-(aq) \rightleftharpoons [CuCl_4]^{2-}(aq) + H_2O(l)$$

$$K_4 = 3.1 \ dm^3mol^{-1}$$

$$[Cu(H_2O)_4]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CuCl_4]^{2-}(aq) + 4H_2O(l)$$

tetraaquacopper(II) ion **tetrachloridocuprate(II) ion**

$$K_f = K_1 \times K_2 \times K_3 \times K_4 = 4.2 \times 10^5 \ dm^{12}mol^{-4}$$

- The overall K_f of $[Cu(NH_3)_4]^{2+}(aq)$ is *larger* than that for $[CuCl_4]^{2-}(aq)$. It shows that NH_3 is a *stronger ligand*, and $[Cu(NH_3)_4]^{2+}(aq)$ is *more stable* than $[CuCl_4]^{2-}(aq)$

Labile Complex Ions

- Complex ions in which ligands can be interchanged rapidly are said to be **labile**. $[Cu(H_2O)_4]^{2+}$, $[Cu(NH_3)_4]^{2+}$, and $[CuCl_4]^{2-}$ are all labile.



Carey B. Van Loon

Labile complex ions
The exchange of ligands in the coordination sphere of Cu^{2+} occurs very rapidly. The solution at the extreme left is formed by dissolving $CuSO_4$ in concentrated HCl(aq). Its yellow color is due to $[CuCl_4]^{2-}$. When a small amount of water is added, the mixture of $[Cu(H_2O)_4]^{2+}$ and $[CuCl_4]^{2-}$ ions produces a yellow-green color. When $CuSO_4$ is dissolved in water, a light blue solution of $[Cu(H_2O)_4]^{2+}$ forms. NH_3 molecules readily displace H_2O molecules as ligands and produce deep blue $[Cu(NH_3)_4]^{2+}$ (extreme right).

Effect of Temperature in Complex-Ion Equilibria

❖ Temperature can be a factor to alter the equilibrium constant in complex ions.

Example: Equilibrium system between $[Co(H_2O)_6]^{2+}$ and $[CoCl_4]^{2-}$





$$[Co(H_2O)_6]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CoCl_4]^{2-}(aq) + 6H_2O(l)$$

hexaaquacobalt(II) ion (pink) tetrachlorocobaltate(II) ion (blue)

Predict

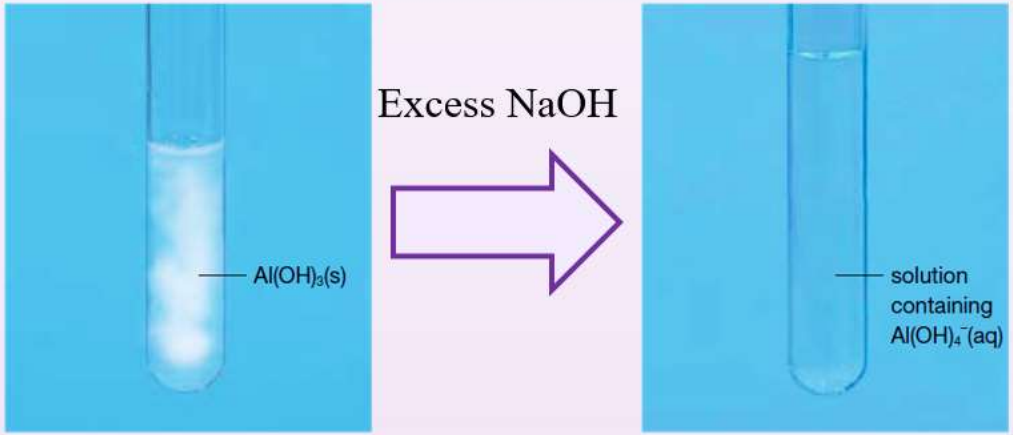
(i) the sign of ΔH of the above equilibrium.

(ii) the changes of equilibrium constant (K) by temperature.

Formation of Complex Ions from Metal Hydroxides

- Most metal hydroxides are **insoluble** in water.
- Some metal hydroxides ***redissolve*** when excess sodium hydroxide or ammonia solution is added to them. It is due to the formation of complex ions.

1. Metal hydroxides in excess NaOH (*aq*)



Examples: (colourless)

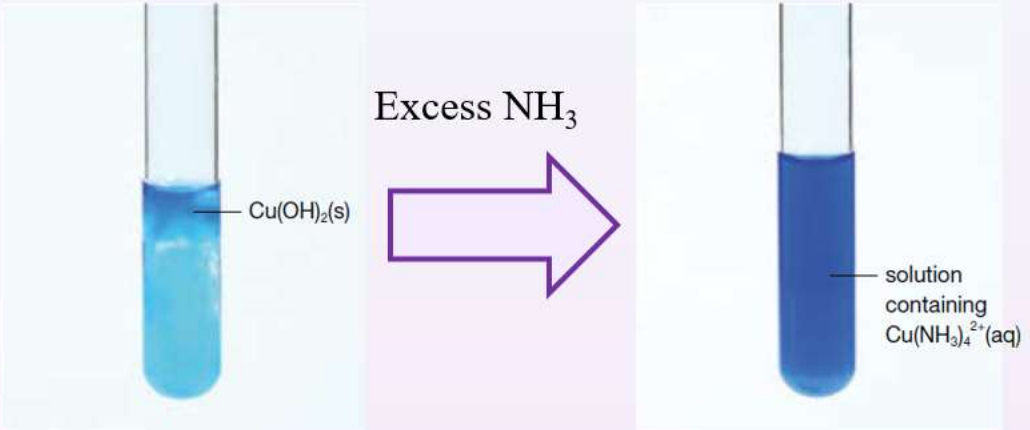
$Al(OH)_3(s) + OH^-(aq) \rightarrow Al(OH)_4^-(aq)$ tetrahydroxoaluminate anion

$Zn(OH)_2(s) + OH^-(aq) \rightarrow Zn(OH)_4^-(aq)$ tetrahydroxozincate anion

$Pb(OH)_2(s) + OH^-(aq) \rightarrow Pb(OH)_4^-(aq)$ tetrahydroxoplumbate(II) anion

Formation of Complex Ions from Metal Hydroxides

2. Metal hydroxides / oxides in excess NH_3 (*aq*)



Examples:

(colourless) tetraamminezinc cation

$$Zn(OH)_2\,(s) + 4NH_3\,(aq) \rightarrow Zn(NH_3)_4^{2+}(aq) + 2OH^-(aq)$$

(deep blue) tetraamminecopper(II) cation

$$Cu(OH)_2\,(s) + 4NH_3\,(aq) \rightarrow Cu(NH_3)_4^{2+}(aq) + 2OH^-(aq)$$

(colourless) diamminesilver(I) cation

$$Ag_2O\,(s) + H_2O\,(l) + 4NH_3\,(aq) \rightarrow 2Ag(NH_3)_2^{+}(aq) + 2OH^-(aq)$$

*Important: **NOT ALL** metal hydroxides could redissolve, it depends on the ***formation (stability) constant*** (K_f) of the corresponding complex ions.

Test for Copper (II) Ions

- To make the **$Cu^{2+}(aq)$** more visible, it is converted to the **deep blue complex (tetraamminecopper(II)) ion $[Cu(NH_3)_4]^{2+}(aq)$** in a reaction in which NH_3 molecules replace H_2O molecules in a complex ion.



Complex-ion formation: A test for $Cu^{2+}(aq)$

$$[Cu(H_2O)_4]^{2+}(aq) + 4\,NH_3(aq) \longrightarrow [Cu(NH_3)_4]^{2+}(aq) + 4\,H_2O(l)$$

(pale blue) **(deep blue)**

- The **deep blue colour** is detectable at much lower concentrations than the **pale blue**; the formation of $[Cu(NH_3)_4]^{2+}$ is a sensitive test for the presence of Cu^{2+}.

Formation Constants for Some Complex Ions

Equilibrium	K_f ((mol dm^{-3})$^{-n}$)
$Cr(OH)_3(aq) + OH^-(aq) \rightleftharpoons [Cr(OH)_4]^-(aq)$	8×10^{29}
$[Fe(H_2O)_6]^{2+}(aq) + 6CN^-(aq) \rightleftharpoons [Fe(CN)_6]^{4-}(aq) + 6H_2O(l)$	$\sim 1 \times 10^{37}$
$[Fe(H_2O)_6]^{3+}(aq) + 6CN^-(aq) \rightleftharpoons [Fe(CN)_6]^{3-}(aq) + 6H_2O(l)$	$\sim 1 \times 10^{42}$
$[Fe(H_2O)_4]^{3+}(aq) + 4Cl^-(aq) \rightleftharpoons [FeCl_4]^-(aq) + 4H_2O(l)$	8×10^{-2}
$[Co(H_2O)_6]^{2+}(aq) + 6NH_3(aq) \rightleftharpoons [Co(NH_3)_6]^{2+}(aq) + 6H_2O(l)$	1.3×10^4
$[Co(H_2O)_6]^{3+}(aq) + 6NH_3(aq) \rightleftharpoons [Co(NH_3)_6]^{3+}(aq) + 6H_2O(l)$	4.5×10^{33}
$[Co(H_2O)_6]^{2+}(aq) + 6CN^-(aq) \rightleftharpoons [Co(CN)_6]^{4-}(aq) + 6H_2O(l)$	$\sim 1 \times 10^{19}$
$[Co(H_2O)_6]^{3+}(aq) + 6CN^-(aq) \rightleftharpoons [Co(CN)_6]^{3-}(aq) + 6H_2O(l)$	$\sim 1 \times 10^{63}$
$[Co(H_2O)_6]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CoCl_4]^{2-}(aq) + 4H_2O(l)$	8.8×10^{-10}
$[Ni(H_2O)_6]^{2+}(aq) + 6NH_3(aq) \rightleftharpoons [Ni(NH_3)_6]^{2+}(aq) + 6H_2O(l)$	5.3×10^8
$[Ni(H_2O)_4]^{2+}(aq) + 4CN^-(aq) \rightleftharpoons [Ni(CN)_4]^{2-}(aq) + 4H_2O(l)$	2×10^{31}
$[Cu(H_2O)_4]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CuCl_4]^{2-}(aq) + 4H_2O(l)$	4.8×10^5
$[Cu(H_2O)_4]^{2+}(aq) + 4CN^-(aq) \rightleftharpoons [Cu(CN)_4]^{2-}(aq) + 4H_2O(l)$	1×10^{25}
$[Cu(H_2O)_4]^{2+}(aq) + 4NH_3(aq) \rightleftharpoons [Cu(NH_3)_4]^{2+}(aq) + 4H_2O(l)$	1.1×10^{13}
$[Zn(H_2O)_4]^{2+}(aq) + 4NH_3(aq) \rightleftharpoons [Zn(NH_3)_4]^{2+}(aq) + 4H_2O(l)$	4.1×10^8
$[Zn(H_2O)_4]^{2+}(aq) + 4CN^-(aq) \rightleftharpoons [Zn(CN)_4]^{2-}(aq) + 4H_2O(l)$	1×10^{18}
$[Zn(H_2O)_4]^{2+}(aq) + 4OH^-(aq) \rightleftharpoons [Zn(OH)_4]^{2-}(aq) + 4H_2O(l)$	4.6×10^{17}
$[Ag(H_2O)_2]^+(aq) + 2NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+(aq) + 2H_2O(l)$	1.6×10^7
$[Ag(H_2O)_2]^+(aq) + 2CN^-(aq) \rightleftharpoons [Ag(CN)_2]^-(aq) + 2H_2O(l)$	5.6×10^{18}

Formation Constants for Some Complex Ions

Stepwise and Overall Formation (Stability) Constants (K_f) for Some Complex Ions

Metal Ion	Ligand	K_1	K_2	K_3	K_4	K_5	K_6	K_f
Ag^+	NH_3	2.0×10^3	7.9×10^3					1.6×10^7
Zn^{2+}	NH_3	3.9×10^2	2.1×10^2	1.0×10^2	5.0×10^1			4.1×10^8
Cu^{2+}	NH_3	1.9×10^4	3.9×10^3	1.0×10^3	1.5×10^2			1.1×10^{13}
Ni^{2+}	NH_3	6.3×10^2	1.7×10^2	5.4×10^1	1.5×10^1	5.6	1.1	5.3×10^8
Cu^{2+}	en	5.2×10^{10}	2.0×10^9					1.0×10^{20}
Ni^{2+}	en	3.3×10^7	1.9×10^6	1.8×10^4				1.1×10^{18}
Ni^{2+}	EDTA	4.2×10^{18}						4.2×10^{18}

- Unlike most other equilibrium constants previously considered, K_f values are usually large numbers. With large K values, it is sometimes convenient to solve a problem in ***two*** steps:

1. assume that the forward reaction goes to completion;
2. assume that a small change occurs in the reverse direction to establish the equilibrium.

Example 10

Determining Whether a Precipitate Will Form in a Solution Containing Complex Ions

A 0.10 mol sample of $AgNO_3$ is dissolved in 1.00 L of 1.00 M NH_3. If 0.010 mol NaCl is added to this solution, will AgCl(s) precipitate?

Given $K_f\,[Ag(NH_3)_2]^+ = 1.6 \times 10^7$

Suggested Solution:

Assuming the formation reaction goes to completion, we obtain

	$Ag^+(aq)$	+ $2\,NH_3(aq)$	⟶ $[Ag(NH_3)_2]^+(aq)$
initial concns, M:	0.10	1.00	
changes, M:	−0.10	−0.20	+0.10
after reaction, M:	≈0	0.80	0.10
initial concns, M:	0	0.80	0.10
changes, M:	$+x$	$+2x$	$-x$
equil concns, M:	x	$0.80 + 2x$	$0.10 - x$

When substituting into the following expression, we make the assumption that $x \ll 0.10$, which we will find to be the case.

$$\frac{[[Ag(NH_3)_2]^+]}{[Ag^+][NH_3]^2} = \frac{0.10 - x}{x(0.80 + 2x)^2} \approx \frac{0.10}{x(0.80)^2} = 1.6 \times 10^7$$

$$x = [Ag^+] = \frac{0.10}{(1.6 \times 10^7)(0.80)^2} = 9.8 \times 10^{-9}\,M$$

$$Q_{sp} = (9.8 \times 10^{-9})(1.0 \times 10^{-2}) = 9.8 \times 10^{-11} < 1.8 \times 10^{-10}$$

AgCl will not precipitate.



Class Practice 10

Determining Whether a Precipitate Will Form in a Solution Containing Complex Ions

Will AgCl(s) precipitate from 1.50 L of a solution that is 0.100 M $AgNO_3$ and 0.225 M NH_3 if 1.00 mL of 3.50 M NaCl is added? Given that K_f of $[Ag(NH_3)_2]^+ = 1.6 \times 10^7$.



Class Practice 11

Controlling a Concentration to Cause or Prevent Precipitation from a Solution of Complex Ions

What is the *minimum* concentration of NH_3 needed to prevent AgCl (s) from precipitating from 1.00 L of a solution containing 0.10 mol $AgNO_3$ and 0.010 mol NaCl?
(Given that K_f of $[Ag(NH_3)_2]^+ = 1.6 \times 10^7$ and K_{sp} of AgCl $= 1.8 \times 10^{-10}$.)



Class Practice 12

Determining the Solubility of a Solute When Complex Ions Form

What is the molar solubility of AgCl in 0.100 M NH_3 (aq)?
(Given that K_f of $[Ag(NH_3)_2]^+ = 1.6 \times 10^7$ and K_{sp} of AgCl = 1.8×10^{-10}.)

Class Practice 13

Suppose 0.100 mol of $AgNO_3$ is dissolved in 1.00 L of a 1.00 M NH_3 (aq). Calculate the concentrations of Ag^+ and $[Ag(NH_3)]^+$ ions present at equilibrium.

Given the following:

(step 1) $Ag^+(aq) + NH_3\,(aq) \rightleftharpoons Ag(NH_3)^+\,(aq)$ $\qquad \dfrac{[Ag(NH_3)^+]}{[Ag^+][NH_3]} = K_1 = 2.1 \times 10^3$

(step 2) $Ag(NH_3)^+(aq) + NH_3(aq) \rightleftharpoons Ag(NH_3)_2{}^+\,(aq)$ $\qquad \dfrac{[Ag(NH_3)_2{}^+]}{[Ag(NH_3)^+][NH_3]} = K_2 = 8.2 \times 10^3$



Class Practice 13 (Continued)

Amphoterism

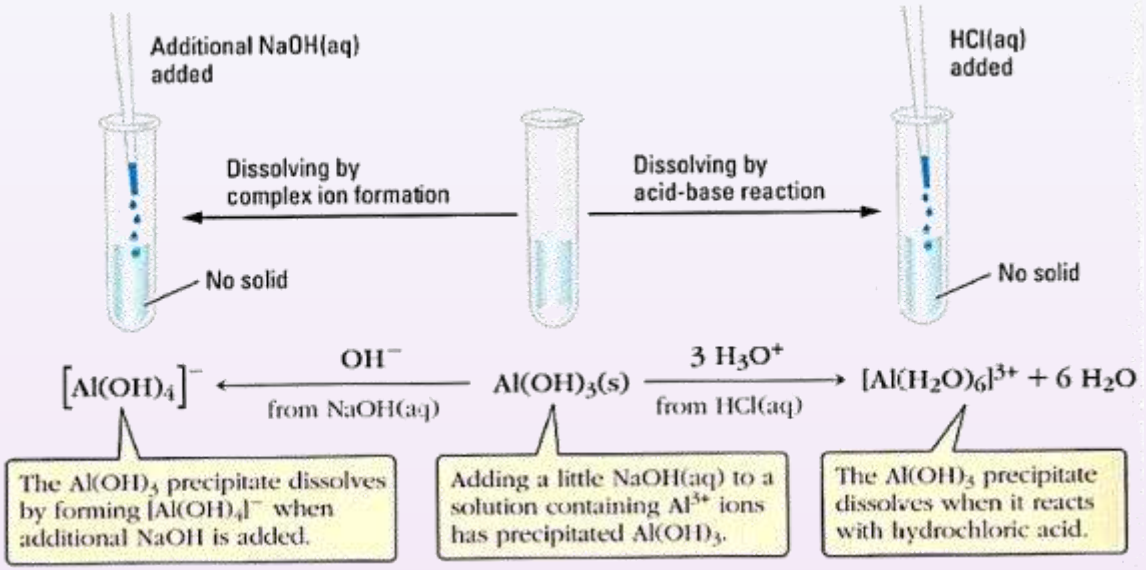
- Some metal oxides and hydroxides that are relatively insoluble in water dissolve in **strongly acidic** and **strongly basic** solutions.
- These substances are soluble in strong acids and bases because they themselves can serve as either an acid or base, i.e. they are ***amphoteric***.
- Amphoteric oxides and hydroxides include those of **Al^{3+}, Cr^{3+}, Zn^{2+}, and Sn^{2+}**.
- Consider **aluminium hydroxide, $Al(OH)_3$**, it is soluble *both* in strongly acidic and in strongly basic solutions.

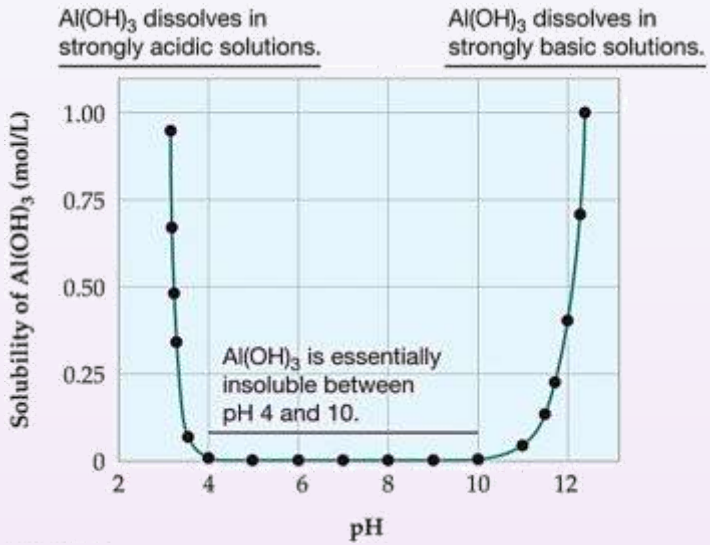
In acid: $Al(OH)_3(s) + 3\ H_3O^+(aq) \rightleftharpoons Al^{3+}(aq) + 6\ H_2O(l)$

In base: $Al(OH)_3(s) + OH^-(aq) \rightleftharpoons Al(OH)_4^-(aq)$

Amphoterism

The amphoteric nature of $Al(OH)_3$





- Amphoteric species dissolve in acidic solutions because they contain basic anions.
- What makes amphoteric oxides and hydroxides special is that they also dissolve in strongly basic solutions. This behaviour results from the formation of **complex anions** containing several hydroxides bound to the metal ion.

$$Al(OH)_3\,(s) + OH^-\,(aq) \rightleftharpoons Al(OH)_4^-\,(aq)$$

Amphoterism

The amphoteric nature of $Al(OH)_3$

- $Al^{3+}(aq)$ is more accurately represented as **$Al(H_2O)_6^{3+}(aq)$.** As a strong base is added, $Al(H_2O)_6^{3+}(aq)$ loses protons in a stepwise manner, eventually forming neutral and water-insoluble **$Al(H_2O)_3(OH)_3$**. This substance then dissolves upon removal of an additional proton to form the anion **$[Al(H_2O)_2(OH)_4]^-$**.

$$Al(H_2O)_6^{3+}(aq) + OH^-(aq) \rightleftharpoons Al(H_2O)_5(OH)^{2+}(aq) + H_2O(l)$$
$$Al(H_2O)_5(OH)^{2+}(aq) + OH^-(aq) \rightleftharpoons Al(H_2O)_4(OH)_2^{+}(aq) + H_2O(l)$$
$$Al(H_2O)_4(OH)_2^{+}(aq) + OH^-(aq) \rightleftharpoons Al(H_2O)_3(OH)_3(s) + H_2O(l)$$
$$Al(H_2O)_3(OH)_3(s) + OH^-(aq) \rightleftharpoons Al(H_2O)_2(OH)_4^{-}(aq) + H_2O(l)$$

- Similar stepwise manner for other amphoteric ions including **$[Cr(H_2O)_6]^{3+}(aq)$ to $[Cr(H_2O)_2(OH)_4]^-$**, **$[Zn(H_2O)_4]^{2+}(aq)$ to $[Zn(OH)_4]^-$**, and **$[Sn(H_2O)_3]^{2+}(aq)$ to $[Sn(OH)_3]^-$**.

Qualitative Cation Analysis

- In ***qualitative analysis***, we determine what substances are present in a mixture but not their quantities.

- An analysis that aims at identifying the cations present in a mixture is called ***qualitative cation analysis***, which provides us with many examples of precipitation (and dissolution) equilibria, acid base equilibria, and redox reactions.

- In the scheme, cations could be divided into 5 groups:

Ag^+, Pb^{2+}, Hg_2^{2+}

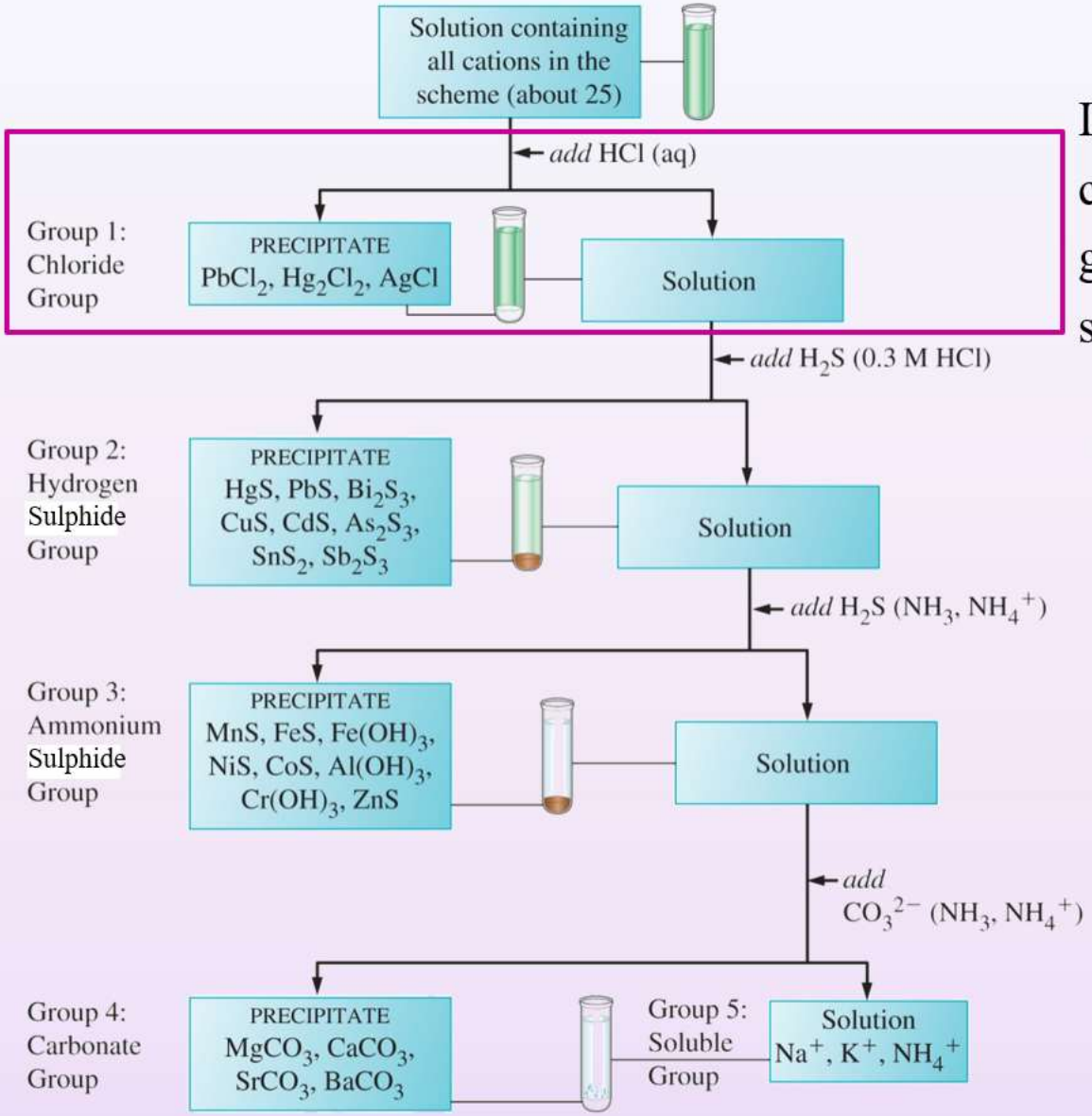
Cu^{2+}, Bi^{3+}, Cd^{2+}, Pb^{2+}, Hg^{2+}, As^{3+}, Sb^{3+}, Sn^{3+}, Sn^{4+}

Al^{3+}, Fe^{2+}, Fe^{3+}, Co^{2+}, Ni^{2+}, Cr^{3+}, Zn^{2+}, Mn^{2+}

Ba^{2+}, Ca^{2+}, Mg^{2+}

Na^+, K^+, NH_4^+

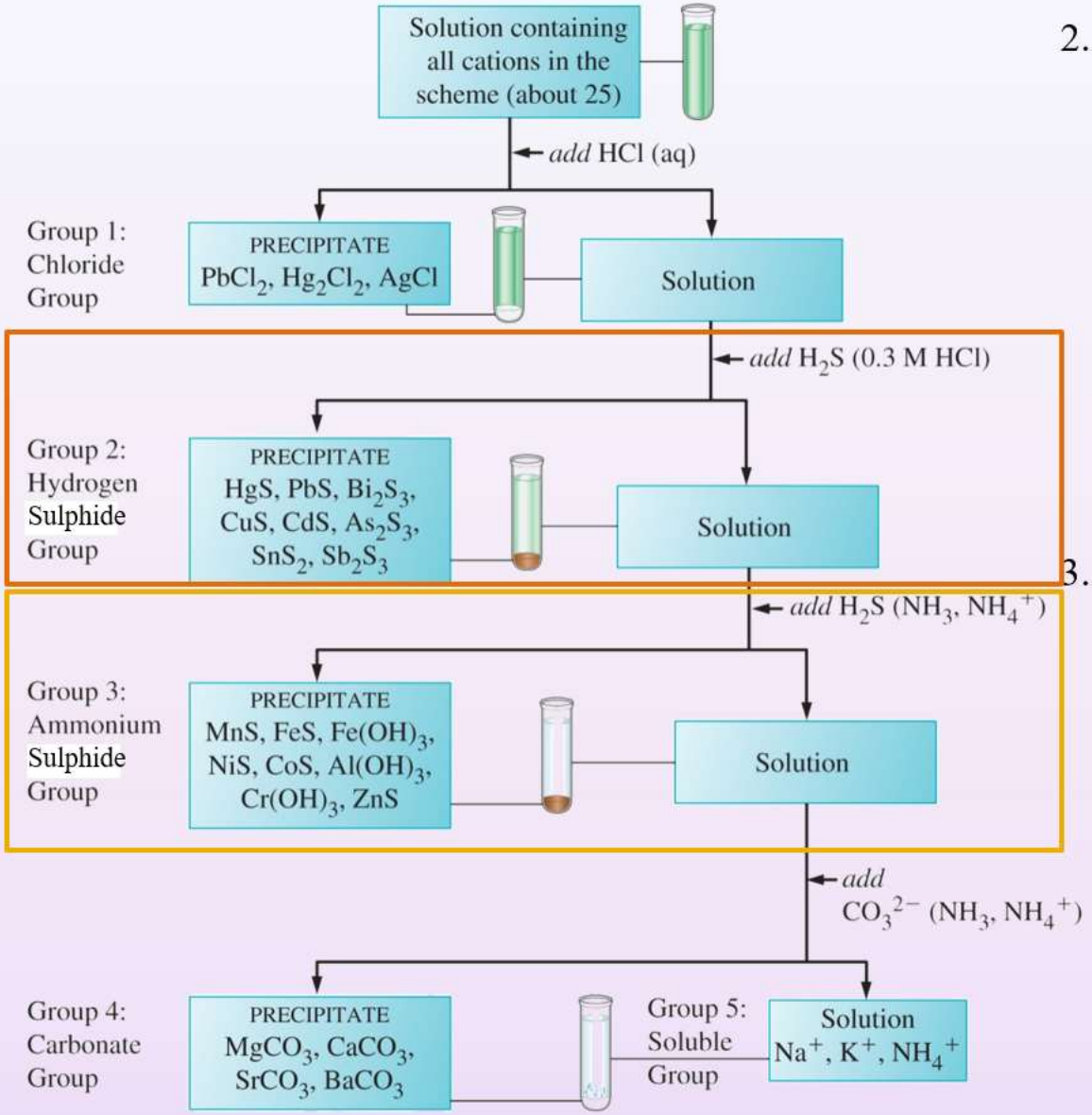
Outline of a Qualitative Cation Analysis



In the scheme, about 25 common cations are divided into *five* groups, depending on differing solubilities of their compounds.

1. The first cations separated, $\mathbf{Pb^{2+}}$, $\mathbf{Hg_2^{2+}}$, and $\mathbf{Ag^+}$ are those with insoluble *chlorides*. The reagent used is ***HCl (aq)***. All other cations remain in solution because their chlorides are soluble.

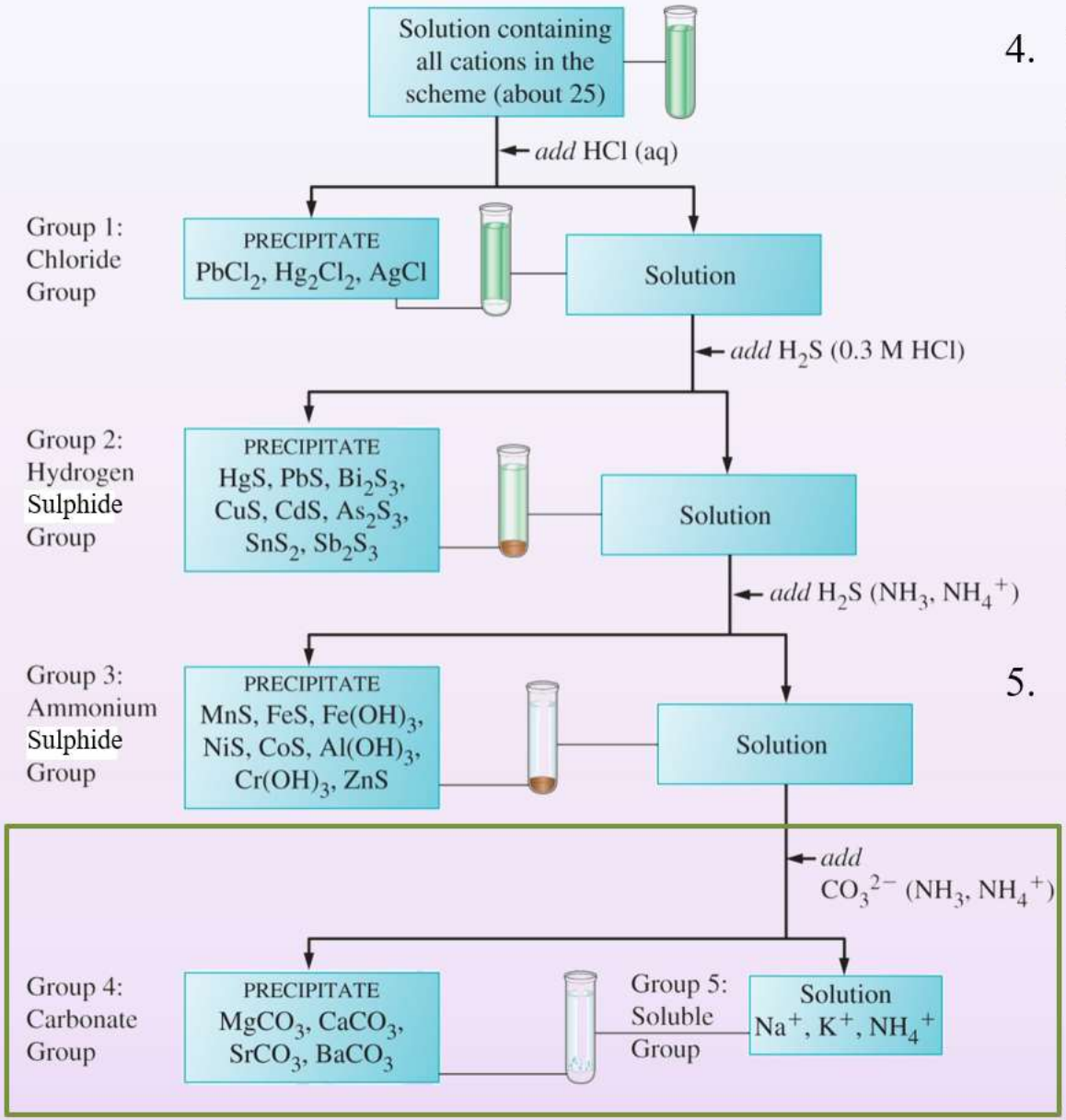
Outline of a Qualitative Cation Analysis



2. After the chloride group precipitate is removed, the remaining solution is treated with ***H_2S in an acidic medium***. Under these conditions, a group of sulphides precipitates. They are known as the hydrogen sulphide group.

3. Next, the solution containing the remaining cations is treated with ***H_2S in a buffer mixture of NH_3 and NH_4^+ ion***, yielding a mixture of insoluble hydroxides and sulfides. This group is called the ammonium sulphide group. The sulphide of Al^{3+} and Cr^{3+} are unstable, reacting with water to form the hydroxides.

Outline of a Qualitative Cation Analysis



4. Treatment of the ammonium sulphide group filtrate with CO_3^{2-} yields the fourth group precipitate, which consists of the carbonates of Mg^{2+}, Ca^{2+}, Sr^{2+}, and Ba^{2+}. It is called the carbonate group because the CO_3^{2-} (aq) is the key reagent.

5. At the end of this series of precipitations, the resulting solution contains only Na^+, K^+ and NH_4^+, and all of whose common salts are water soluble.

Cation Group 1: The Chloride Group

- If a precipitate forms when a solution is treated with HCl(aq), one or more of the following cations must be present: Pb^{2+}, Hg_2^{2+}, and Ag^+.

(a) Test for $PbCl_2$

- Of the three chlorides in the group precipitate, $PbCl_2(s)$ is the most soluble (i.e. larger K_{sp}). When the precipitate is washed with hot water, a sufficient quantity of $PbCl_2$ dissolves to permit a test for Pb^{2+} in the solution. In this test, a lead compound less soluble than $PbCl_2$ such as ***lead(II) chromate***, precipitates.



(hot)

Example: $PbCl_2\,(aq) + Na_2CrO_4\,(aq) \rightarrow \mathbf{PbCrO_4\,(s)} + 2NaCl\,(aq)$

Ionic equation: $Pb^{2+}\,(aq) + CrO_4^{2-}\,(aq) \rightarrow \mathbf{PbCrO_4\,(s)}$

Cation Group 1: The Chloride Group

(b) Test for AgCl and Hg_2Cl_2

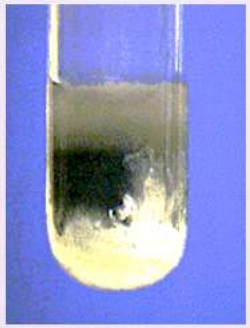
- The portion of the chloride group precipitate that is insoluble in hot water is then ***treated with excess $NH_3(aq)$***.

- **AgCl(s)**: re-dissolves and forms the complex cation $[Ag(NH_3)_2]^+$

$$AgCl(s) + 2\,NH_3(aq) \longrightarrow \underset{\text{diamminesilver(I) cation}}{[Ag(NH_3)_2]^+(aq)} + Cl^-(aq)$$

- When the solution is acidified with **$HNO_3(aq)$**, any silver ion present precipitates as AgCl(s).

- **$Hg_2Cl_2(s)$**: undergoes a redox reaction (***disproportionation***). The products are finely divided black mercury (Hg (l)) and white mercury(II) amidochloride ($HgNH_2Cl$).



$$Hg_2Cl_2(s) + 2\,NH_3(aq) \longrightarrow \underbrace{Hg(l) + HgNH_2Cl(s)}_{\text{Dark gray}} + NH_4^+(aq) + Cl^-(aq)$$

Qualitative analysis **confirmation of mercury(I) ion**

Cation Group 1: The Chloride Group

Summary:

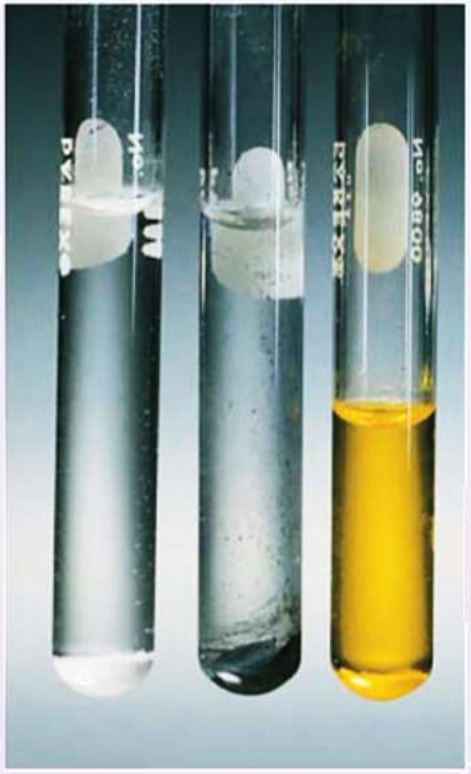
Chloride group precipitates

(a) Group 1 precipitate: Adding HCl to form a mixture of $PbCl_2$ (white), Hg_2Cl_2(white), and AgCl (white).

(b) Test for Hg_2^{2+}: Adding excess NH_3 (aq) to form a mixture of **Hg (black)** and **$HgNH_2Cl$ (white)**.

(c) Test for Pb^{2+}: Adding CrO_4^{2-} (aq) to form a yellow precipitate of **$PbCrO_4$(s)**.

(d) Test for Ag^+: Adding excess NH_3 (aq) to re-dissolve as colourless **$[Ag(NH_3)_2]^+$** , then add H^+ to re-precipitate the **AgCl (s).**



(a) (b) (c)

Cation Groups 2 and 3: Equilibria Involving Hydrogen Sulphide

- Aqueous hydrogen sulphide (hydrosulphuric acid) is the key reagent in the analysis of cation groups 2 and 3. In aqueous solution, H_2S is a weak diprotic acid.

$$H_2S(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + HS^-(aq) \qquad K_{a_1} = 1.0 \times 10^{-7}$$
$$HS^-(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + S^{2-}(aq) \qquad K_{a_2} = 1 \times 10^{-19}$$

- The extremely small value of K_{a2} suggests that S^{2-} is a very strong base, as seen in the following hydrolysis reaction and K_b value.

$$S^{2-}(aq) + H_2O(l) \rightleftharpoons HS^-(aq) + OH^-(aq)$$
$$K_b = K_w/K_{a_2} = (1.0 \times 10^{-14})/(1 \times 10^{-19}) = 1 \times 10^5$$

- Very little S^{2-} can exist in an aqueous solution and that sulphide ion is probably not the precipitating agent for sulphides.
- One way to handle the precipitation and dissolution of sulphide precipitates is to **use acidic solutions**. This approach is reasonable because most sulphide separations are carried out in acidic solution.

Cation Groups 2 and 3: Equilibria Involving Hydrogen Sulphide

Consider

(1) the solubility equilibrium equation for PbS,

(2) an equation for the reverse of the first ionization of H_2S, and

(3) an equation that is the reverse of the self-ionization of water.

Combining these equations to obtain an overall equation that shows the dissolving of PbS(s) in an acidic solution. The equilibrium constant for this overall equation is generally referred to as K_{spa}.

(1)	$PbS(s) + H_2O(l) \rightleftharpoons Pb^{2+}(aq) + HS^-(aq) + OH^-(aq)$	$K_{sp} = 3 \times 10^{-28}$
(2)	$H_3O^+(aq) + HS^-(aq) \rightleftharpoons H_2S(aq) + H_2O(l)$	$1/K_{a_1} = 1/1.0 \times 10^{-7}$
(3)	$H_3O^+(aq) + OH^-(aq) \rightleftharpoons H_2O(l) + H_2O(l)$	$1/K_w = 1/1.0 \times 10^{-14}$
Overall:	$PbS(s) + 2\,H_3O^+(aq) \rightleftharpoons Pb^{2+}(aq) + H_2S(aq) + 2\,H_2O(l)$	$K_{spa} = ?$

$$K_{spa} = \frac{K_{sp}}{K_{a_1} \times K_w} = \frac{3 \times 10^{-28}}{(1.0 \times 10^{-7})(1.0 \times 10^{-14})} = 3 \times 10^{-7}$$

Example 11

Separating Metal Ions by Selective Precipitation of Metal Sulphides

Show that PbS(s) will precipitate but FeS(s) will not precipitate from a solution that is 0.010 M in Pb^{2+}, 0.010 M in Fe^{2+}, saturated in H_2S (0.10 M H_2S), and maintained with $[H_3O^+] = 0.30$ M. For PbS, $K_{spa} = 3 \times 10^{-7}$; for FeS, $K_{spa} = 6 \times 10^{2}$.

Suggested Solution:

$$MS(s) + 2\,H_3O^+(aq) \rightleftharpoons M^{2+}(aq) + H_2S(aq) + 2\,H_2O(l)$$

Calculating Q_{spa} we have

$$Q_{spa} = \frac{[M^{2+}][H_2S]}{[H_3O^+]^2} = \frac{0.010 \times 0.10}{(0.30)^2} = 1.1 \times 10^{-2}$$

For PbS: Q_{spa} of $1.1 \times 10^{-2} > K_{spa}$ of 3×10^{-7}. PbS(s) will precipitate.

For FeS: Q_{spa} of $1.1 \times 10^{-2} < K_{spa}$ of 6×10^{2}. FeS(s) will not precipitate.



Class Practice 14

Separating Metal Ions by Selective Precipitation of Metal Sulfides

Show that $Ag_2S(s)$ ($K_{spa} = 6 \times 10^{-30}$) should precipitate and that FeS(s) ($K_{spa} = 6 \times 10^{2}$) should not precipitate from a solution that is 0.010 M Ag^+ and 0.020 M Fe^{2+}, and maintained with $[H_3O^+] = 0.30$ M.

Cation Groups 2 and 3: Equilibria Involving Hydrogen Sulphide

- In the qualitative cation analysis, it is necessary to both **precipitate** and **redissolve** sulphides.

- **Methods of dissolving metal sulphides:**

1. To react the metal sulphide with **an acid**

- According to Le Châtelier's principle, the solubility increases as the solution is made more acidic – equilibrium is shifted to the right.

$$MS(s) + 2\,H_3O^+(aq) \rightleftharpoons M^{2+}(aq) + H_2S(aq) + 2\,H_2O(l)$$

- As a result, some water-insoluble sulphides, **Cation Group 3 (e.g. MnS, FeS, CoS, NiS, ZnS)** (larger K_{spa}), are readily soluble in strongly acidic solutions. If an NH_3 buffer is applied to remove the acid, they will **re-precipitate** again.
- **Cation Group 2 (e.g. PbS, CuS and HgS)** (smaller K_{spa}), cannot be dissolved in acidic solutions because their K_{sp} values are too low.

Cation Groups 2 and 3: Equilibria Involving Hydrogen Sulphide

2. To use an ***oxidizing*** acid such as $HNO_3(aq)$.

❖ The sulphide ion is oxidized to elemental **sulphur (S)** and the free metal ion appears in solution, as in the dissolving of CuS(s).

$$3\,CuS(s) + 8\,H^+(aq) + 2\,NO_3^-(aq) \longrightarrow 3\,Cu^{2+}(aq) + 3\,S(s) + 2\,NO(g) + 4\,H_2O(l)$$

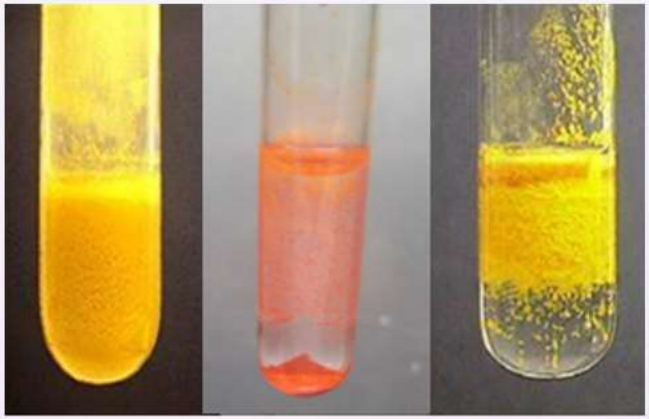
Separation of cation Group 2

❖ A few metal sulphides dissolve in a basic solution with a high concentration of HS^-, just as acidic oxides dissolve in solutions with a high concentration of OH^-.

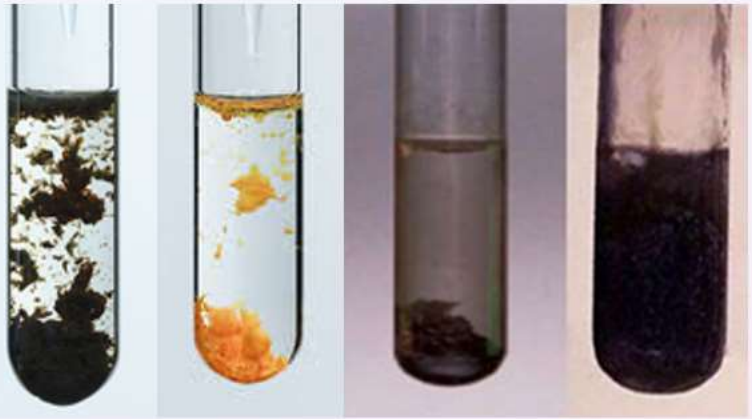
❖ This property is used to advantage in separating the eight sulphides of **cation group 2, the hydrogen sulphide group**, into two subgroups. The subgroup consisting of **HgS, PbS, CuS, CdS, and Bi_2S_3 remains unchanged** after treatment with an alkaline solution with an excess of HS^-, but **As_2S_3, SnS_2, and Sb_2S_3 dissolve**.

Cation Groups 2 and 3: The Sulphide Group

Some appearance of sulphides (cation Group 2 and 3) and hydroxides (Group 3)



SnS_2 Sb_2S_3 As_2S_3



CuS CdS PbS Bi_2S_3



Cinnabar (HgS)

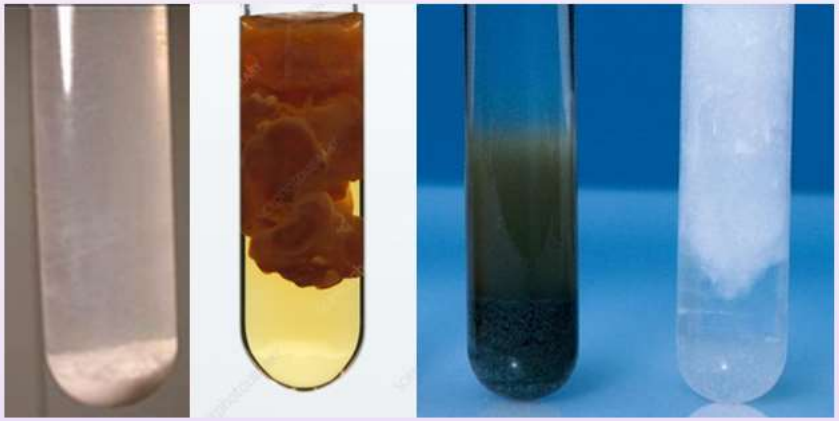


MnS

FeS

CoS

NiS



ZnS $Fe(OH)_3$ $Cr(OH)_3$ $Al(OH)_3$

Cation Groups 2 and 3: The Sulphide Group

The K_{sp} of some metal sulphides

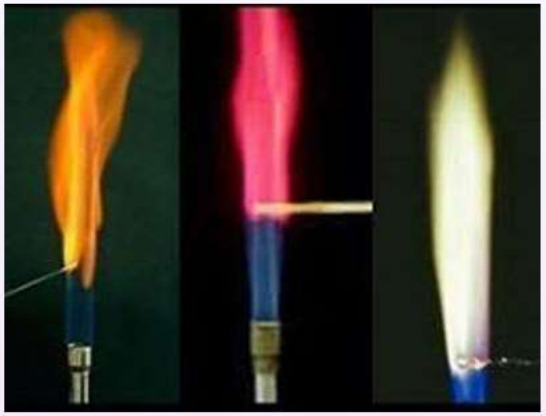
Metal Sulphides	K_{sp}
MnS	2.5×10^{-13}
FeS	6.3×10^{-18}
NiS	3.1×10^{-19}
CoS	4.0×10^{-21}
ZnS	1×10^{-23}
SnS	1×10^{-25}
CdS	7.9×10^{-27}
PbS	1×10^{-27}
CuS	6.3×10^{-36}
HgS	4.0×10^{-53}
SnS_2	1×10^{-70}

Cation Group 4: The Carbonate Group

Carbonates of some Group II elements by precipitation:

$MgCO_3$
$CaCO_3$
$SrCO_3$
$BaCO_3$

Test:



Ca^{2+} Sr^{2+} Ba^{2+}

Cation Group 5: The Soluble Group

Cations that remain soluble after precipitation:

K^+, Na^+, NH_4^+

Test:





K^+ Na^+ NH_4^+



Class Practice 15

Both Cu^{2+} and Ag^{+} are present in the same aqueous solution. Explain which of the following reagents would work best in separating these ions, precipitating one and leaving the other in solution: $(NH_4)_2CO_3(aq)$, $HNO_3(aq)$, $H_2S(aq)$, HCl(aq), $NH_3(aq)$, or NaOH(aq).

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